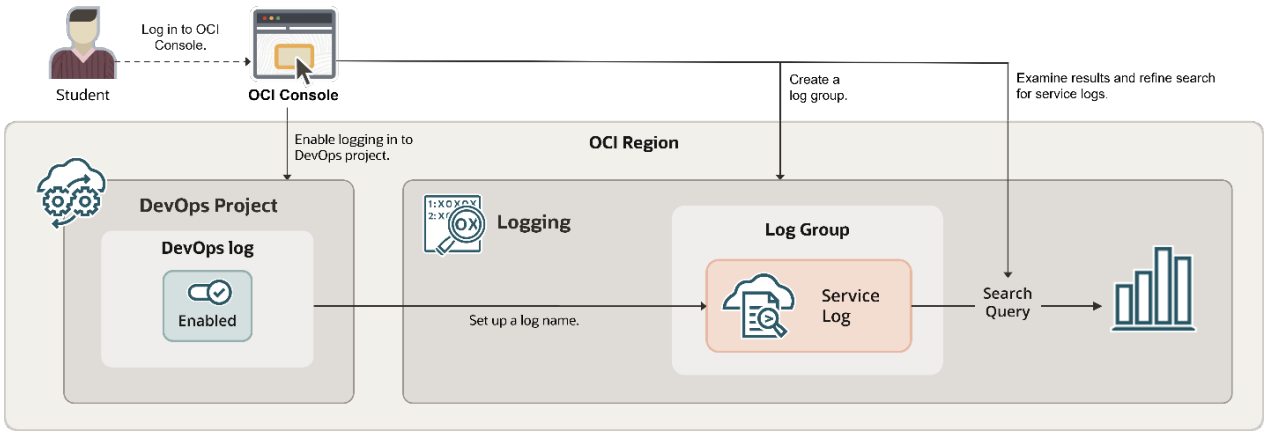
Lab 16. Manage DevOps Project Log Using OCI Console

Overview

The Oracle Cloud Infrastructure Logging service offers a fully managed, highly scalable single point of access to all the logs in your tenancy. Logging provides access to logs from Oracle Cloud Infrastructure resources. These logs include critical diagnostic information that describes how resources are performing and being accessed.



In this lab, you'll:

a. Configure logs for DevOps project
b. Run the build manually
c. Search your logs

Prerequisites

- You must complete the following labs before you perform tasks for this practice:
  - *Create Docker image for a web application using Dockerfile*.
  - *Create and work with OCIR repository*.
  - *Set up cluster access*
  - *Deploy a sample Web application on a cluster using kubectl*.
  - *Work with code repositories in OCI DevOps project*.
  - *Create and set up artifacts and environments in DevOps project*.
  - *Automate Web App deployment to an OKE cluster using OCI DevOps CI/CD pipeline*.

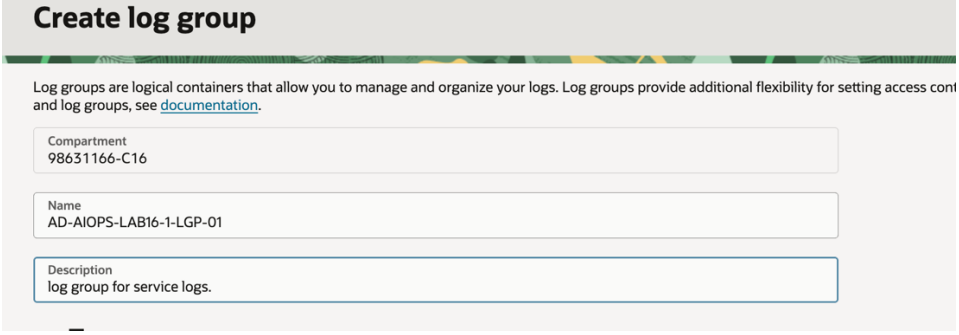
Configure Logs for DevOps Project

The log groups are logical containers for organizing and managing logs. A log must always be inside a log group. You will first create a log group to enable or create logs.

After creating a log group, you will update logging for the DevOps Project.

1. Open the navigation menu and select **Observability & Management**. Under **Logging**, click **Log Groups**.
2. Select your `<assigned compartment>` from List scope on the left menu.

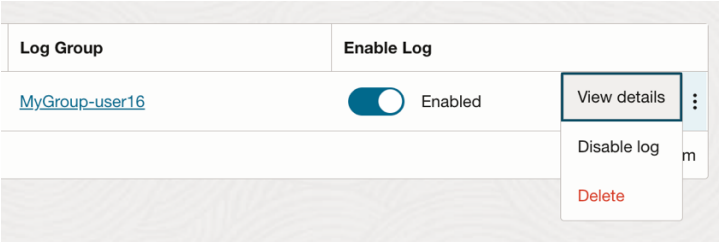
3. Click **Create Log Group**.
4. In the **Create Log Group** window, enter the following values:
   - **Compartment**: Use Applied filter to select your `<assigned-compartment>`.
   - **Name**: `IAD-AIOPS-LAB16-1-LGP-01`
   - **Description**: `log group for service logs.`



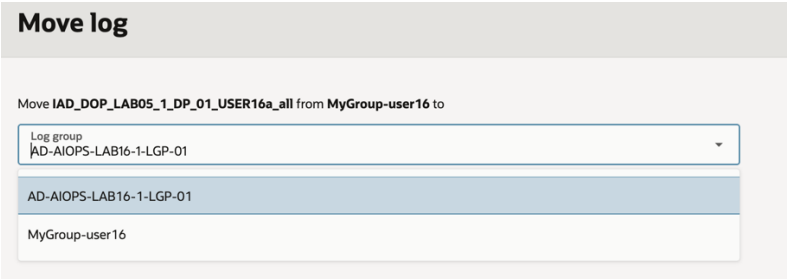
5. Click **Create**.

   The DevOps Project is created with logs enabled with a default log group. You will change this log group to **IAD-AIOPS-LAB16-1-LGP-01** and create a new log for DevOps Logs.

6. From **Developer Services**. Under **DevOps**, click **Projects**.
7. Open the DevOps project **IAD-DOP-LAB05-1-DP-01-<*userID*>**.
8. Under **DevOps project resources** section on left menu, click **Logs**.
9. Click the three dots on the right to open the Actions menu and click view details.

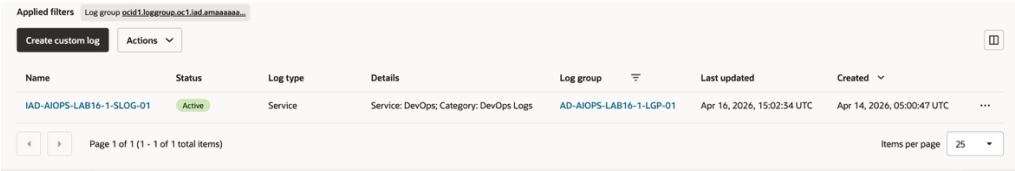


10. From the **Actions** drop on right, click **Change Log Group**.



   a. Click the drop down for log group and select **IAD-AIOPS-LAB14-1-LGP-01** from the drop-down list.
   b. Click **Move Resource** from bottom right.
   c. From the **Actions** drop on right, click **Edit**
   d. Enter the Log Name as `IAD-AIOPS-LAB16-1-SLOG-01`
   e. Click **Update**.

11. Open the navigation menu and select **Observability & Management**. Under **Logging**, click **Log Groups**.
12. Select your `<assigned compartment>` from Applied filters.
13. Click Log Group **IAD-AIOPS-LAB16-1-LGP-01.**
14. Click **Logs**, verify the log name with `IAD-AIOPS-LAB16-1-SLOG-01` is shown **Active** for **Service: DevOps; Category: DevOps Logs**.



Run the Build

Execute the build pipeline which will trigger the deployment pipeline as well. During and after the execution for build and deployment pipelines, the logs are generated.

1. Open the **DevOps project IAD-DOP-LAB05-1-DP-01-*<userID>***.
2. Select **Build Pipelines** on the left menu and click **IAD-DOP-LAB11-1-BPL-01.**
3. Verify that three stages are available in the build pipeline: Build-Demo-WebApp, Push WebApp Artifacts, and Trigger OKE Deployment.
4. Click **Start manual run** on the top right corner of the page.
5. Keep **Build run name** as default and click **Start manual run**. The **Status** on the top left will be shown as **In progress**, and the execution will take approximately 5 minutes to complete.
6. Upon completion, the status on top left will be updated to **Succeeded**.
7. Click your DevOps project **IAD-DOP-LAB05-1-DP-01-*<userID>*** using the breadcrumb list at the top of the page and click **Deployments** from the left menu.
8. Verify if the status of the last deployment is **Succeeded**.

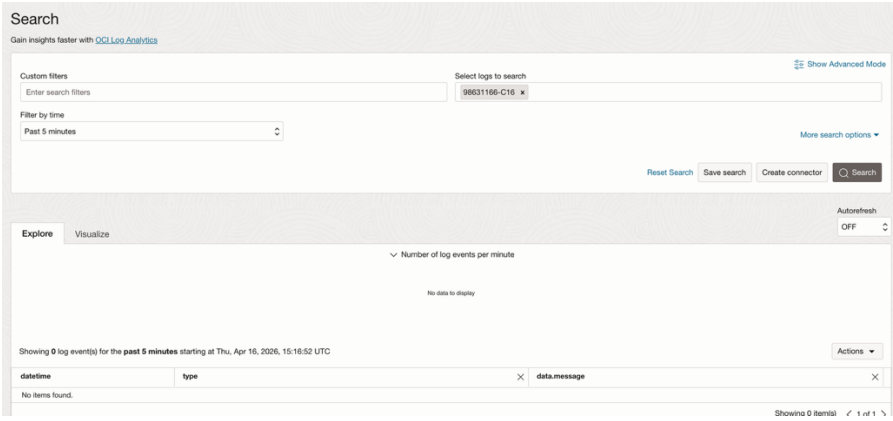
Search Your Logs

You will explore the contents of your logs and become familiar with the built-in search capabilities provided by the logging service. You will learn to select service logs to be included in search, examine results, and refine search for service logs. Additionally, you will update a build stage and execute build run to generate corresponding log records.

You will also learn to search logs from saved searches.

Search your logs

1. Open the navigation menu and select **Observability & Management**. Under **Logging**, click **Search.**
2. Click **Select logs to search** text field. The **Select logs to search** window appears.
3. In the **Select logs to search** field, click **(x)** to remove your `<assigned_compartment>` if selected by default.

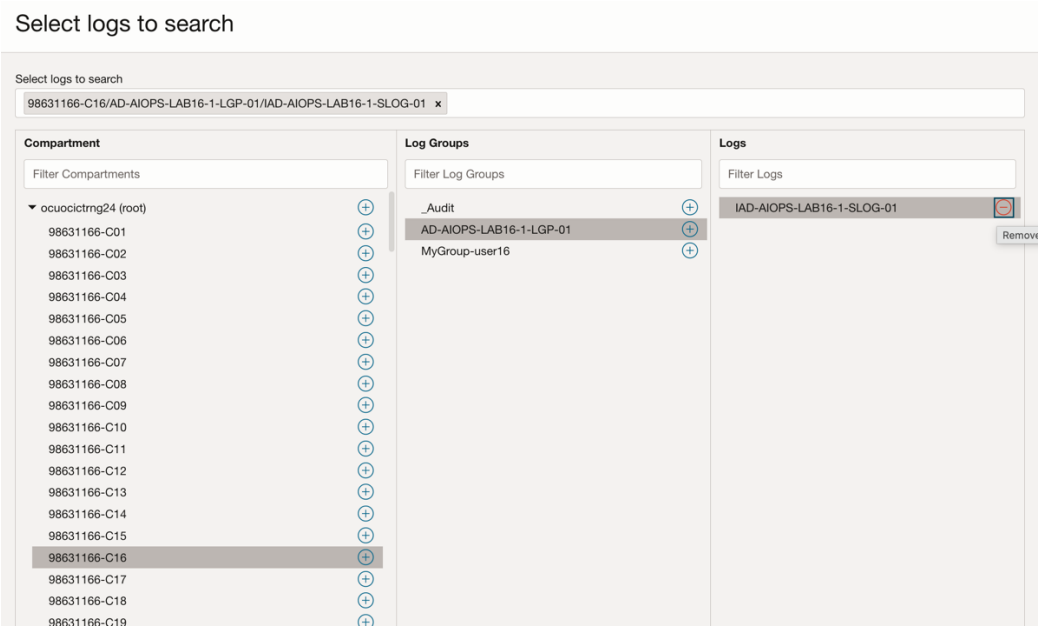


4. Expand the root compartment under the **Compartment** column and select your *<assigned_compartment>* from the Compartment list. Do not click the plus **(+)** sign. Click the compartment name only.

   This will bring up the log groups in that compartment without including the compartment itself as part of the search criteria. You don't want the compartment itself included, because you don't want all the logs for that compartment in the search results.

5. In the **Log Groups** column, select **IAD-AIOPS-LAB16-1-LGP-01** log group, but again, **click the name only** without clicking the plus **(+)** sign. This will bring up the logs for that log group.

   In the **Logs** column, select **IAD-AIOPS-LAB16-1-SLOG-01** log. This time, click the plus **(+)** sign to add it as the only search criteria. The **Select logs to search** field at the top of the window will be updated to *<assigned_compartment>*/IAD-AIOPS-LAB16-1-LGP-01/IAD-AIOPS-LAB16-1-SLOG-01.



6. Click **Continue** to execute the search.

Explore Filters

1. To examine results and refine search for service logs, in the **Custom filters** field at the top of the Search area, enter the following and hit enter on your keyboard:

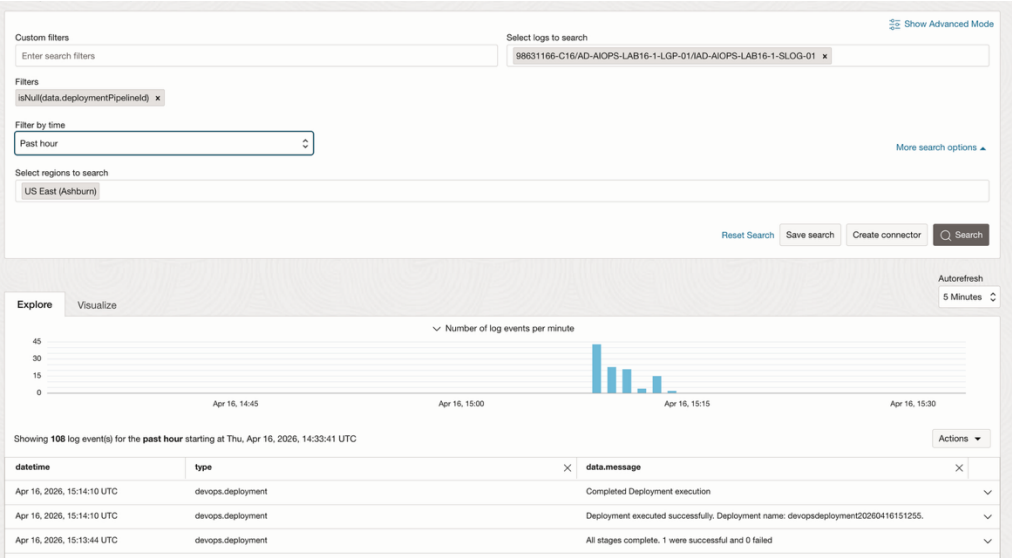
   `data.deployPipelineId = <Select OCID from the list> or null`

   This will show the log records that were executed for the selected pipeline.

2. If there are no log records displayed, then **Filter by time** and select **Today** from the drop-down list. This will return all the log records matching the condition for the entire day.
3. In the **Custom filters** field at the top of the Search area, enter the following and hit enter on your keyboard:

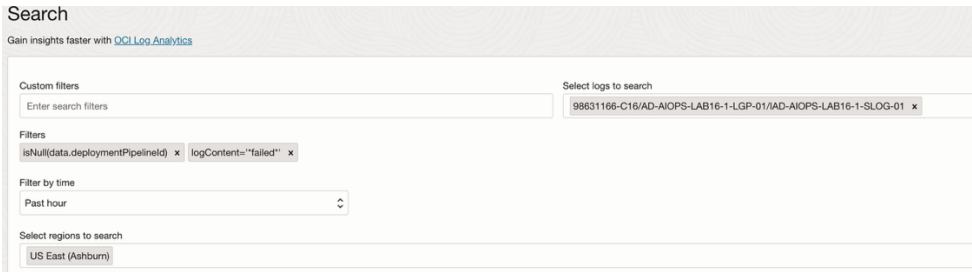
   `data.message = Completed Deployment execution`

   This will show the log records that contains the message **Completed Deployment execution** for the selected pipeline. You can remove the search filters by clicking **(x)** for **Filters** under **Custom filters**.



4. In the **Custom filters**, enter the keyword `failed` and hit Enter on your keyboard.

   The **Filters** will show the following filter applied, which returns log records that contain the keyword `failed: logContent='*failed*'`



5. Select **Filter by time** as **Past 5 minutes**. Verify there is no recent log data. You may note the timestamp if there are log records.
6. Click **Save search**. Enter the following values in the form:
   - **Search Name**: `Deployment-failures`

- **Compartment**: Select your `<assigned compartment>`.
- **Description**: `Search for failed deployments.`
- Click **Save Search**.

7. Click **Reset Search** to reset the search filters.

Update a build stage and generate corresponding log records

1. Open the DevOps project **IAD-DOP-LAB05-1-DP-01-<*userID*>**.
2. Select **Build Pipelines** on the left menu and click **IAD-DOP-LAB11-1-BPL-01.**
3. Click the three dots on **Trigger OKE Deployment** and click **View details**.
4. Click **Edit Stage** and **uncheck** to disable the **Send build pipelines Parameters** box.

   Note: This will block the build pipeline parameters to be shared with deployment pipeline and thus trigger a failure in deployment.

5. Click **Save changes**.
6. Click **Start manual run**.
7. Keep the build run name as default and click **Start manual run**.
8. Wait until the status of build run shows **Succeeded**.
9. Click your DevOps project **IAD-DOP-LAB05-1-DP-01-<*userID*>** using the breadcrumb list at the top of the page and click **Deployments** from the left menu.
10. Verify that the last deployment is shown as **Failed**.
11. From the navigation menu, select **Observability & Management**, then click **Search** under **Logging.** Under Logging, click **Saved Searches** to reach previously saved search `Deployment-failures`.
12. Choose **Filter by time** as **Past 15 minutes** if the data is not shown for **Past 5 minutes**. Verify the log records containing the string `failed` is shown for the deployment failures.
13. Expand one of the log records by clicking the down-arrow icon on the right.
    a. On the **JSON** tab, the log data is shown in JSON format. Review the **message** under the **data** section, which shows the log message for the failure.
    b. Click the **Before & After** Tab. This shows the logs representing what was going on before and after the log message was generated, which helps in the troubleshooting.
14. From the OCI Console **Main Menu**, select **Developer Services**. Under **DevOps**, select **Projects**.
15. Select your `<assigned compartment>` from List scope on the left menu.
16. Click Project **IAD-DOP-LAB05-1-DP-01-<*userID*>** under Project name column.
17. Select **Build Pipelines** on the left and click the pipeline **IAD-DOP-LAB11-1-BPL-01.**
18. Click the three dots on Trigger OKE Deployment stage. Click **View details.**
19. Click **Edit Stage**, enable **Send build pipelines Parameters.**
20. Click **Save changes**, Do NOT click **Start manual run**.
21. Click **DevOps Projects** on top to return to the Projects page.

Congratulations! You have successfully configured and explored logs for your build and deployment pipelines in your DevOps Project.

Purge Instructions

Purge instructions for Logs and Log Groups

1. Open the navigation menu and select **Observability and Management**. Under **Logging**, click **Log Groups**.
2. Click **IAD-AIOPS-LAB16-1-LGP-01** log group.
3. Under **Resources** in the left menu, click **Logs**.

4. For **IAD-AIOPS-LAB16-1-SLOG-01**, click the three dots on the right to open the Actions menu and click **Delete**. Confirm to Delete.
5. Click **Log Group**.
6. For **IAD-AIOPS-LAB16-1-LGP-01,** click the three dots on the right to open the Actions menu and click **Delete**. Confirm to Delete.